Auteur: Siebe Haché
Studierichting: Informatica
Oefeningenreeks 3D nr. 8.6

$$f(x) = \frac{x^3}{(x+2)^2}$$

$$f'(x) = \frac{3x^2(x+2)^2 - x^3 \cdot 2(x+2)}{(x+2)^4} = \frac{3x^2(x+2) - 2x^3}{(x+2)^3} = \frac{x^2(x+6)}{(x+2)^3}$$

$$f''(x) = \frac{(3x^2+12x)(x+2)^3 - (x^3+6x^2) \cdot 3(x+2)^2}{(x+2)^6} = \frac{24x}{(x+2)^4}$$

$$\frac{24x}{(x+2)^4} = 0 \Rightarrow x = 0$$

x		-2		0	
$f''(x)$	$-$	$\vert^{(4)}$	$-$	0	$+$
$f(x)$	$\frown$	$\vert^{(2)}$	$\frown$	B	$\smile$

$$f(0) = \frac{0^3}{(0+2)^2} = 0$$

$$f'(0) = \frac{0^2(0+6)}{(0+2)^3} = 0$$

$$y - 0 = 0(x-0) \Rightarrow y = 0$$

References